# Michael Kamensky

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## Education

**B.S. Computer Engineering**, University of California, Santa Cruz **3.60 GPA**
**Relevant Coursework:** Advanced Multivariate Calculus, Linear Algebra, Computer Architecture, Data Structures and Algorithms, Programming Abstractions: Python, Computer Systems, Discrete Mathematics, Parallelism and Concurrency, Advanced Programming, Computer Networks, Network Security, Circuit Analysis

## Experience

**Marchup Internship(3 months):** Contributed to efforts to horizontally scale a social media service at MarchUp by re-architecting a web-server-hosted search engine deployed on a monolithic PHP/Apache stack. Designed and evaluated multiple AWS-based distributed prototypes, including a real-time indexing architecture that improved search freshness and scalability.
**Reference:** Harsh Parandekar (Boss/Manager) **Email(best):** harsh@marchup.net **Phone:** 408-230-9028

**Dymium Internship(8 months):** Implemented a scalable Kubernetes architecture for a stateful reverse tunneling proxy using a custom DNS-based load-balancing protocol, enabling horizontal scaling of previously single-instance servers.
**Reference:** Igor Plotnikov(Boss/Manager) **Email(best):**igor@dymium.io **Phone:** 408-702-7865

**QUIC Research University of California, Santa Cruz (6 months):** Conducted research under Professor Ram Sundara Raman in network systems and security, developing a QUIC protocol fuzzer to evaluate firewall behavior. Designed packet mutation strategies to identify evasion and non-evasion cases, systematically documenting results, and co-authored a paper analyzing fuzzing results on open-source firewalls.
**Reference:** Ram Sundara Raman (Professor) **Email(best):** rsundar2@ucsc.edu

## Projects

**QUIC Fuzzer**     https://github.com/r-andlab/mutate-quic
A network security fuzzing framework designed to evaluate firewall behavior on QUIC traffic. The system generates and captures valid QUIC packets, applies protocol-aware mutation strategies, and replays modified traffic against firewall-protected targets, recording and analyzing evasion and non-evasion outcomes to characterize firewall handling of encrypted transport protocols. Findings were systematically analyzed and synthesized into written research artifacts, emphasizing critical evaluation of firewall behavior and protocol handling.
https://github.com/michaelkamensky/Fuzzing-Firewalls-with-QUIC/blob/main/cse195_senior_thesis.pdf

**Custom DNS server**
A scalable reverse tunneling system built on Kubernetes that replaces single-instance proxies with a horizontally scalable design. The architecture employs a custom DNS server deployed as a Kubernetes Service and propagated to workloads via ConfigMaps, enabling dynamic traffic routing to active tunneling proxies through a DNS-based load-balancing protocol using zero-TTL records and proxy-reported state via API calls.

## Skills

**Primary Focus (Production & Research Experience)**

- C (4+ years) — systems programming, networking, embedded systems
- Python (6+ years) — scripting, analysis, tooling
- Go (1+ year) — distributed systems, networking tools

**Networking & Security**

- QUIC protocol analysis (6 months)
- Protocol fuzzing & firewall evaluation (6 months)
- Network security & encrypted transport analysis (coursework + research)

**Distributed Systems & Infrastructure**

- Kubernetes (8 months) — stateful services, DNS-based routing, scaling
- AWS (3 months) — distributed prototypes, backend services
- DNS-based load balancing (project experience)

**Additional Languages & Technologies**

- Java (coursework + projects)
- C++ (coursework)
- RISC-V Assembly (coursework + labs)
- PHP (project experience)
- HTML/CSS/JavaScript (supporting knowledge)